

TFW 18E COURSE

THEORETICAL

TFW-TACSOP-1.1

TFW JTAC Cheatsheet

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Module 1: Communications Sergeant

1 Responsibilities

1.1 Radio Systems

Within ACRE most radio systems work very similarly. Meaning that if you are able to use the AN/PRC 152 you should be able to use the AN/PRC-117F, especially since they are made by/for the same country. These are all the radio systems explained:

- AN/PRC-117F
 - 20W transmit power
 - 30MHz - 512MHz range
 - 20+ km range
 - American made
- AN/PRC-152
 - 5W transmit power
 - 30MHz - 512MHz range
 - 5+ km range
 - American made
- AN/PRC-148
 - 5W transmit power
 - 30MHz - 512MHz
 - 5-7 km range
 - American made (not in use)
- AN/PRC-343
 - 100mW transmit power
 - 2.4GHz - 2.483GHz
 - 850 m range
 - American made
- AN/PRC-77
 - 4W transmit power
 - 30MHz - 80MHz
 - 3-5 km range
 - American made
- Beofeng 888S
 - 5W transmit power
 - 0.4kHz - 0.47kHz
 - 3-5 km range
 - Made in China
- SEM 52 SL
 - 1W transmit power
 - 46kHz - 65.975MHz
 - 2-4 km range
 - German made
- SEM 70
 - 4W transmit power
 - 46kHz - 65.975MHz

- 3-5 km range
- German made
- WS No. 38
 - 200mW
 - 7.4MHz - 9MHz
 - 800 m range
 - Made in the United Kingdom

1.2 Communications

- Communications
- Channel assignments
- JTAC
- HLZ

1.3 Encryption codes

TFAR uses three separate encryption codes to identify which radios can communicate to each other. These are: West, East and Independent codes. Meaning that if you will not be able to communicate with radios that have different encryption codes. Backpack based radios will also have these codes.

- **West encryption codes:**
 - AN/PRC-152
 - Persistent Systems MPU-5
 - RF-7800S-TR
- **East encryption codes:**
 - FADAK
 - PNR-1000A
- **Independent encryption codes:**
 - AN/PRC-148 JEM
 - AN/PRC-154

Module 2: Brevity

Reference the [brevity](#)

2.1 NATO Standard Phonetic Alphabet & Numbers

In radio communications, "brevity" refers to the practice of conveying information as clearly and concisely as possible. This is crucial to avoid misunderstandings and ensure efficient use of airtime. Brevity involves using

standard phrases, abbreviations, and procedures to reduce the length of transmissions while maintaining clarity and accuracy. It helps in minimising radio traffic and allows for quicker communication, especially in critical situations.

NATO Phonetic Alphabet			
A	Alpha	N	November
B	Bravo	O	Oscar
C	Charlie	P	Papa
D	Delta	Q	Quebec
E	Echo	R	Romeo
F	Foxtrot	S	Sierra
G	Golf	T	Tango
H	Hotel	U	Uniform
I	India	V	Victor
J	Juliet	W	Whiskey
K	Kilo	X	X-ray
L	Lima	Y	Yankee
M	Mike	Z	Zulu

Number	Code Word	Phonetic Pronunciation
0	Zero	ZEE-RO
1	One	WUN
2	Two	TOO
3	Three	TREE
4	Four	FOW-ER
5	Five	FIFE
6	Six	SIX
7	Seven	SEV-EN
8	Eight	AIT
9	Nine	NIN-ER
100	Hundred	HUN-DRED
1000	Thousand	TOU-SAND

Module 3: Radio etiquette

3.1 Introduction to radio etiquette

Brevity and Radio Etiquette

Effective radio communication is crucial in operational environments, where miscommunication can lead to critical errors. To minimise the risk of misunderstandings, it is essential to adhere to established brevity and radio etiquette rules. These guidelines are designed to ensure that information is conveyed clearly and concisely, using standard phrases and abbreviations. This approach helps to reduce airtime and prevent unnecessary radio traffic, enabling swift and accurate exchanges.

Operational Considerations for JTACs

Joint Terminal Attack Controllers (JTACs) must balance their role in radio communication with their fundamental duties as riflemen. In combat situations, adhering to the **Shoot, Move, Communicate** principle is vital:

- 1) **Shoot:** Prioritise direct engagement and the safety of your team.
- 2) **Move:** Ensure tactical positioning and movement are maintained.
- 3) **Communicate:** Use radio communication effectively, but as a secondary priority to immediate combat actions.

Practical Tips

- **Positioning:** Whenever possible, position yourself towards the rear during operations. This allows you to focus on communications without compromising your ability to respond to immediate threats.
- **Prioritisation:** Remember that effective radio communication should not override basic combat responsibilities. Prioritise actions based on the situation's demands, ensuring that communication supports, rather than hinders, operational effectiveness.

Following these guidelines helps maintain smooth and efficient communication, ensuring mission success and the safety of all personnel.

3.2 Basic radio etiquette

1) Think before you speak:

- a) Plan your message before pressing the push-to-talk (PTT) button. This helps in delivering a concise and clear message without unnecessary pauses or filler words.

2) Use proper protocol:

- a) Begin communications with a call sign, identifying yourself and the intended recipient. For example, "Alpha Team, this is Bravo Team, over."
- b) Acknowledge all messages with "Roger" or "Copy" to confirm receipt and understanding.

3) Speak clearly and calmly:

- a) Speak slowly, clearly, and in a calm, even tone. Avoid shouting or mumbling.
- b) Articulate your words properly to ensure they are easily understood, especially under noisy or stressful conditions.

4) Keep it short and concise:

- a) Use short, concise sentences. Long or overly detailed messages can lead to misunderstandings and are harder to follow.
- b) Use standard phrases and terminology relevant to your context (e.g. "Over", "Out", "Roger", "Say again")

5) Avoid interruptions:

- a) Wait for a natural pause or the end of a transmission before speaking. Ensure the channel is clear before you transmit.
- b) If urgent communication is needed, use a priority call or phrase, like "Break, break" to indicate an emergency or urgent message.

6) Use common phrases and terms

- a) **"Over"**: End of transmission; expecting a reply.
- b) **"Out"**: End of transmission; no reply expected.
- c) **"Roger"**: Message received and understood.
- d) **"Copy"**: Acknowledges the receipt of the message.
- e) **"Wilco"**: Will comply; indicates understanding and intent to comply.
- f) **"Say Again"**: Request to repeat the last message.
- g) **"Break"**: Separates different parts of a message.
- h) **"Affirmative"** and **"Negative"**: Yes and no, respectively, used to avoid confusion.

7) Avoid using personal names:

- a) Use call signs or role identifiers instead of personal names. This maintains operational security and clarity.

8) Avoid unauthorised frequencies:

- a) Use only the channels or frequencies authorised for your communication. Unauthorised use can disrupt other communications.

9) Listen first:

- a) Before transmitting, listen to ensure the frequency is not in use. This prevents interrupting ongoing communications.

10) Maintain professionalism:

- a) Avoid unnecessary chatter, slang, or humour on the radio. Keep communications professional and focused on the task at hand.

11) End communications properly:

- a) Conclude conversations with "Out" or another appropriate sign-off to indicate the end of communication.

12) Acknowledge all messages:

- a) Acknowledge the receipt of all messages, even if only to confirm understanding. This helps to ensure the sender knows their message was received.

13) Practise good security:

- a) Be mindful of the sensitive nature of information shared over the radio, especially in military and law enforcement settings. Avoid disclosing classified or sensitive information.

3.3 All stations check

An all stations check (sometimes called: “sound off”) should be performed before step off into any operational environment as well as after any major communications changes.

The all stations check is initiated by a leadership member or by the communications sergeant, it is then performed based on seniority. The 18E is responsible for anything regarding radios. Therefore the comms sergeant needs to ensure everyone's radio works before step off.

All stations check for ground personnel is in this order:

- 1) Detachment Commanders (18A/180A)**
- 2) Operations Sergeant (18Z)**
- 3) Intelligence Sergeant (18F)**
- 4) Weapons Sergeant (18B)**
- 5) Engineer Sergeant (18C)**
- 6) Medical Sergeant (18D)**
- 7) Communications Sergeant (18E)**
- 8) Candidates**

Everyone sounds off with their callsign. Some scenarios might change the order. When doing military freefall jumps the sound off will go by number assigned. Leadership or the communications sergeant can change the order as appropriate.

Module 4: Communications Sergeant responsibilities

4.1 Mission required equipment

The Communications Sergeant (18E) should evaluate the pros and cons of each communications system to determine their necessity and identify any resupply needs. During pre-operation checks, the sergeant must ensure all radios are operational, set the correct frequencies, and coordinate closely with the 18Z to select the appropriate equipment.

Key considerations include ensuring seamless communication with air, artillery, and other friendly assets. The 18E must also confirm that the 18A can communicate with other ground units, the 18B with artillery assets, and the 18Z with all detachment members.

4.2 Fall back plans

Ensure there is a secondary communications system set up. Such as using the ATAK messaging system or signalling with smokes, flares or chemlights. Or using SATCOM as an alternative. It is important to have answers to these questions to ensure mission success. If something fails you need a backup plan.

To do this we use PACE:

- **Primary:** The main form of communication.
- **Alternate:** Used if the primary method fails. Is often nearly as reliable as the primary method.
- **Contingency:** Third option, used if both the primary and alternate methods fail. But it is often still somewhat of a reliable form of communication.
- **Emergency:** Last resort, is less secure or slower. Might be visual or audible signals.

4.3 Pro words & other pre set brevity terms

Often shown in the opord. However, it is still important to confirm or make the pro words. Main things to think about are:

- **Theme:** Such as themes for HLZs, BPs and HPs.
- Phase Lines (18Zs responsibility)
- **Pro Words:** Codewords made to help with brevity over comms to signal mission milestones.
- **Landmarks:** Mark landmarks and name them for quick reference in a stressful situation. Such as: Roads, Buildings or hills.

The reason you want to use this is to help with brevity and to ensure secure comms.

4.4 Planning and briefing

When planning, the communications sergeant needs to take a lot of things into account. The following 8 points are important to have a plan for.

1) Mission Analysis and Understanding Intent

- **Objectives and Intent:** Understanding the overall mission objectives and the commander's intent.
- **Friendly Forces:** Knowing the disposition, capabilities, and planned movements of friendly forces.
- **Enemy Forces:** Gathering intelligence on enemy positions, capabilities, and likely courses of action.

2) Target Development

- **Target Identification:** Determining high-value and priority targets that need to be engaged.

- **Target Location:** Accurately locating targets, including coordinates, descriptions, and identifying features.
- **Target Prioritization:** Establishing which targets should be engaged first based on mission priorities and threat levels.

3) Coordination and Integration

- **Coordination with Ground Forces:** Ensuring synchronisation with ground commanders and units, including understanding their plans and communication protocols.
- **Airspace Coordination:** Deconflicting airspace to avoid conflicts between different air assets and ground-based fire.
- **Communications Plan:** Establishing communication methods, frequencies, and procedures with aircraft and ground units.

4) Air Support Planning

- **Allocation of Air Assets:** Determining the types and numbers of aircraft required, including fighters, bombers, reconnaissance, and unmanned aerial vehicles (UAV).
- **CAS Requests:** Submitting and coordinating requests for CAS, specifying the type of support needed, such as strikes or shows of force.
- **Weapon Selection:** Choosing appropriate munitions based on target type, collateral damage concerns, and desired effects.

5) Fire Support Coordination

- **Control Measures:** Establishing boundaries, no-fire areas, and restricted fire areas to protect friendly forces and civilians.

6) Risk Assessment and Mitigation

- **Collateral Damage Estimates (CDE):** Evaluating potential collateral damage and planning to minimise it.
- **Rules of Engagement (ROE):** Understanding and adhering to the applicable ROE to ensure lawful and ethical use of force.

7) Briefings and Rehearsals

- **Pre-Mission Briefings:** Conducting briefings with pilots, ground commanders, and other stakeholders to ensure everyone understands the plan and contingencies.
- **Rehearsals:** Participating in rehearsals or simulations to practise the execution of the plan and refine coordination. (If possible)

8) Equipment and Logistics

- **Equipment Checks:** Ensuring that all communication equipment, targeting tools, and personal gear are functional and ready for use.
- **Logistics Coordination:** Arranging for necessary supplies, such as batteries, maps, and targeting devices.

4.5 Joint Terminal Attack Controller

It will be the communications sergeants job to coordinate with air assets to assist ground troops. This includes: setting up landings zones, providing Close Air Support (CAS) and providing safety for ground and air assets. A JTAC's job is to:

- 1) **Coordinate Air Support:** they will coordinate with ground forces and pilots to ensure that air support is integrated effectively into ground operations.
- 2) **Targeting and Fire Control:** they will identify and prioritise targets, ensuring that munitions hit their intended targets accurately and effectively, while minimising the risk of collateral damage.
- 3) **Communication and Guidance:** JTACs maintain communication with aircraft, ground commanders, and supporting units. They provide pilots with critical information, such as enemy positions, friendly locations, and weather conditions.
- 4) **Safety Assurance:** They are responsible for ensuring the safety of friendly forces by providing accurate information to avoid friendly fire incidents.
- 5) **Battlefield Assessment:** JTACs assess battlefield conditions and adapt their guidance to changing tactical situations, ensuring effective use of air assets.

4.6 Getting accurate grids

There is three ways to get target grids:

- 1) Using the vector 21 connected to the micro dagger to get accurate grids.
 - a) This gives you a 10 digit grid of both your target and yourself.
 - b) It gives precise directions, elevation, distance and speed.
- 2) Using the vector 21 connected to the ATAK.
 - a) This gives you a 10 digit grid as well (providing your cTab settings are set to 1m accuracy).
 - b) It gives directions, elevation and provides blue force tracking (BFT) allowing for a good idea of where friendlies are.
- 3) Using map, compass and map tools.
 - a) It is harder and slower to use while in an operational area.
 - b) Less accurate but a good alternative when the other devices fail.

Module 5: Joint Terminal Attack Controller (JTAC)

5.1 JTAC responsibilities

1) Maintain Currency:

- Ensure proficiency in all air interoperability aspects.
- Stay updated with the latest communication guidelines.

2) Communicate with Ground Force Commander (GFC):

- Keep the GFC informed about available air assets and their capabilities.

3) Mission Planning with GFC and Aviators:

- Develop a plan to effectively use air assets for missions.
- Include the following elements in the plan:
 - Pre-planned strikes
 - Landing Zones (LZs)
 - Drop Zones (DZs)
 - Holding Points (HPs)/Holding Areas (HAs)
 - Battle Positions (BPs)
 - Initial Points (IPs)
 - Any other relevant information

4) Ground Force Plan (GFP):

- Coordinate with the GFC to include the GFP.
- Ensure that aircrews understand the mission's purpose.

5) Further responsibilities during terminal attack control:

Figure below provides minimum responsibilities.

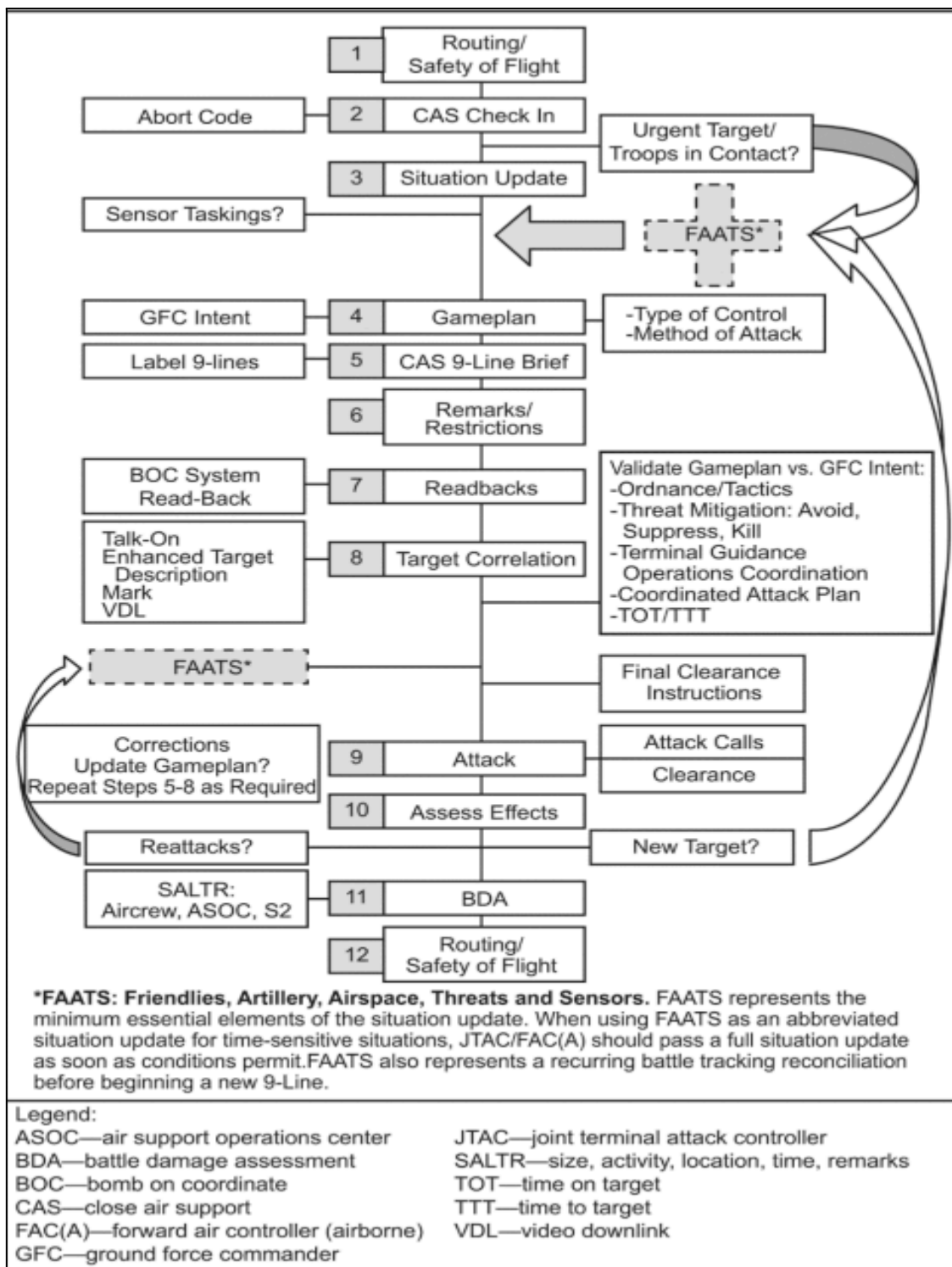
Table 28. Terminal Attack Control Roles and Minimum Responsibilities	
Role	Minimum Responsibilities
Brief	• Check In Brief • Situation Update • CAS Brief (FW+RW A/S fires, S/S fires) • BDA Report • Inflight Report
Stack	• RW Deconfliction • FW Deconfliction • Deconfliction from S/S fires • Deconfliction from A/S fires • Threat mitigation
Mark	• Target correlation • Marks • Terminal guidance operations
Control	• Terminal attack control • Supported Commander Approval Coordination • Attack monitoring
Legend: A/S—air-to-surface BDA—battle damage assessment CAS—close air support	
FW—fixed-wing RW—rotary-wing S/S—surface-to-surface	

5.2 12 Steps of CAS

Reference [12 steps of CAS](#)

The 12 Steps of Close Air Support (CAS) ensure the safety of aircrews and ground forces. JTACs use these steps to quickly guide airframes in identifying and engaging targets.

Figure below provides an example of a CAS execution flow.



5.21 Routing & Safety of Flight

Routing & Safety of Flight is used to inform the aircraft of where you want them to go and any safety concerns. The JTAC will provide this information on first contact with the airframe.

Figure below provides considerations for routing and safety of flight.

Table 29. Routing and Safety of Flight Considerations
• Three-dimensional directions from the current aircraft position to the intended holding point or area. • Holding point and altitude, once established, and which agency to contact. • Other aircraft on station. • Surface-to-air threats that may immediately affect close air support aircraft. • Any other safety of flight issues. • Pass any gun-target-line information (active or inactive) and azimuth/location.

5.22 CAS Check-in Briefing

Reference: [CAS Check-in](#)

Used to inform the JTAC of what the aircrafts capabilities are. Such as CAS, ISR or transport. It will also inform the JTAC of more information of the airframe not previously discussed.

It will include:

- **Callsign of aircraft**
- **Mission number or air tasking order**
- **Number and type of aircraft**
- **Position and altitude**
- **Ordnance**
- **Playtime or time on station (TOS).** Playtime refers to the time the aircraft will be available determined by conditions such as fuel. TOS is often a predetermined time that the aircraft will be available with no conditions.
- **Capabilities.** Such as type of sensors, Video Down Link (VDL) and SITREPs on board.
- **Abort code.** If there is a specific pro word to abort an attack run. If it is default (3x abort) the pilot will either skip it or say that the abort code is: “abort in the clear”.

5.23 Situation Report (SITREP)

JTACs or FAC(A)s will provide the situation update to arriving aircraft, when able. JTACs and FAC(A)s will use the TEFACHR format. The TEFACHR format is also used when doing a Battle Handover (BHO), prior to change in the JTAC or FAC(A), a handover brief shall be conducted with the oncoming controller (Battle Handover).

Figure below provides the sitrep in the TEFACHR format.

Table 31. Situation Update TEFACHR Format		
Line	Situation Update	Battle Handover (BHO)
Threat	•General locations of surface-to-air threats not already covered. •Time of last observed surface-to-air fires.	
Enemy Situation	•General enemy disposition. •Avoid listing grids. Target grids will be addressed in CAS briefs.	•General enemy disposition. •Ground commander's targeting priority list. •Target location grids (may require breaking up transmission). •Ground commander's attack guidance matrix.
Friendly Update	•General friendly situation and scheme of maneuver. •Use geographic references, phase lines, checkpoints, etc. The technique is to use general terms: "all friendlies are east of the 94 easting." •Avoid passing friendly grids. If needed, use no more than 6 digits. •All factor friendly forces during TOS, not just the JTAC. •All CAS assets, ordnance, and TOS remaining for the BHO.	
Artillery	•List factor indirect fire assets. This may include general direction of fire.	•Determine the firing unit's location, call sign, frequency, and status.
Clearance Authority	•Omit this if the speaker has control. •Clarify roles if there are multiple voices on tactical air direction. For example, "Broadsword 11 has control and is located in the combat operations center. My JFO call sign, Mustang, is located with Charlie Company and is up this net." •Define who has authority for brief, stack, mark, and control. •Pass a plan for approving fires for BHO.	
Hazards	•Towers, minimum safe altitude, weather (including surface winds), CBRN, explosives.	
Remarks and Restrictions	Remarks and restrictions may include the following: •Ordnance restrictions. •JTAC capabilities (e.g., laser, infrared, video downlink). •Factor ACM/FSCM. •Radio calls required by controller. •Intent for aircraft (e.g., CAS, multisensory imagery reconnaissance). •Updates to preplanned ACM/FSCM/MCM. •Provide a positive passing of the appropriate elements of brief, stack, mark, and control during BHO. •CBRN targeting restrictions. •Other remarks.	

5.24 CAS Briefs

This is the part where either the 5-line or 9-line will come into play. Pilots only have to read back the target lines and units of measure are always metric. This part will include:

- Target location, description and altitude
- Closest friendly location referenced from target
- Any remarks restrictions

After the brief will follow target correlation. Which will confirm the pilot has the correct target and then the JTAC will tell the pilot to attack.

5.25 Battle Damage Assessment (BDA)

After the attack has been completed the JTAC or pilot (if requested) will provide a BDA. Which will in plain english inform the pilot what the effect on target was. The Battle Damage Assessment should have the following things included: size, activity, location, time, remarks (SALT-R).

Figure below provides an example of the SALT-R format.

Table 33. Battle Damage Estimate (BDA) Report
Size: " _____." (number and type of equipment/personnel observed)
Activity: " _____." (movement direction, stationary, dug-in)
Location: " _____."
Time: " _____."
Remarks: " _____." (munitions expended, observed damage, mission number, and mission accomplished)
Munitions Effective Assessment: " _____."
Re-attack Recommendation: " _____."

5.3 5-Line & 9-Line

Reference [JTAC Cards CAS 9-Line & 5-Line](#)

In Close Air Support (CAS) operations, rotary-wing and fixed-wing aircraft utilise distinct briefing formats due to their differing operational characteristics. Rotary-wing aircraft employ a 5-line brief, which provides essential information, as they can hover and operate at lower altitudes. In contrast, fixed-wing aircraft require a 9-line brief that includes more detailed target descriptions and additional safety protocols. The 5-line format is also applicable for use with the AC-130 Spooky II. These standardised procedures ensure the effective and safe coordination of air assets.

5.31 5-Line in depth

A 5-Line does not require the game-plan to be mentioned separately because it is integrated within the brief.

1) Observer / warning order / game plan

- a) Includes the [type of control](#), [Method Of Attack \(MOA\)](#) and ordnance.

2) Friendly location / mark

- a) Is always referenced from the target position. (Avoid using grids as this will give away the friendly location.
- b) If a mark is available this should also be mentioned in here.

3) Target location

- a) This includes target location in either: grid (if [BOT](#) 6 digit, if [BOC](#) 8 or more)

4) Target description / [mark](#)

- a) General description of target (i.e. BTR-60 or T-90).
- b) Can be marked by: Smoke, IR Pointer, Laser Designator (include laser code) or visual pointer.

5) [Remarks / Restrictions](#)

- a) Final attack heading
- b) Laser Target Line (LTL, this is the direction from where the laser is being pointed) and Pointer Target Line (PTL, this is the direction from where the IR Pointer is pointed)
- c) Surface-to-air threats
- d) Airspace coordination areas (ACAs)
- e) Danger close and initials.
- f) Time on Target (TOT, this is the zulu time for example 1600 zulu) and time to target (TTT, this is the time in hours or minutes for example 0+02 would be 2 minutes)
- g) Post launch abort coordination and considerations

5.32 9-Line in depth

9 Lines require the game plan to be separate from the brief. The game plan includes the [type control](#), [MOA](#), effects desired or ordnance and interval.

9-Line CAS Brief:

1) Initial Point (IP)

- a) Initial Point is where the pilot will start their attack run. This will either be determined in the air briefing before the operation. Or this will be a point determined with the keyhole method.

2) Heading from IP to target.

- a) Can also have an offset. Which would make the pilot go left or right of the heading that has been assigned.

3) Distance from IP to target in meters

4) Target elevation in meters Mean Sea Level (MSL)

- a) Target elevation is important for reference so that the pilot can confirm they are looking at the correct target.

5) Target description

- a) General target description (i.e. BTR-60 or T-90)

6) Target location (6-digit for BOT and 8 digit or more for BOC)

- a) Can also be visual if the pilot already has eyes on.

7) [Type mark / terminal guidance \(is not required but highly recommended\)](#)

8) Location of friendlies relative to target and mark

9) Egress

- a) Use cardinal directions or predetermined points.

Then follow the [remarks / restrictions](#).

5.33 Type of controls

- **Type 1 control:** Used when the JTAC can see both the pilot and the aircraft. Pilot needs to wait for the JTAC to visually confirm the pilot is on the right target before getting the cleared hot. Can only be used in gun or Hydra 70 runs.
- **Type 2 control:** JTAC sees either the target or the aircraft. Typically used when the JTAC only sees the target. Can be used with any type of ordnance.
- **Type 3 control:** JTAC can neither see the target nor the aircraft. Or the aircraft is cleared to engage more than once. The JTAC gives clearance right after target correlation with the term: "Cleared to engage".

5.34 Method Of Attack (MOA)

- **Bomb On Target (BOT)** is used when the JTAC does not have a precise grid but does have visuals to give further plain english target correlation.
- **Bomb On Coordinate (BOC)** is used when the JTAC is able to acquire a 8-digit grid or 10-digit grid. Often used with GPS guided munitions or when the target is not easily identifiable such as a specific tree line.

5.35 Marking methods

IR Pointer Procedure:

- 1) **JTAC:** *Delivers the 5-line brief.*
- 2) **Pilot:** *Read back the 5-line brief.*
- 3) **JTAC:** *"Good readback, say when ready for sparkle."*
- 4) **Pilot:** *"Ready sparkle."*
- 5) **JTAC:** *Points IR pointer at the target, "Sparkle."*
- 6) **Pilot:** *"Contact sparkle, snake."*
- 7) **JTAC:** *Moves IR pointer in a snake pattern, "Snaking."*
- 8) **Pilot:** *"Tally, cease sparkle."*
- 9) **JTAC:** *Stop using the IR pointer, "Ceasing sparkle."*

Important note: the JTAC will listen to the pilots instructions, not the other way around, the goal in the end is for the pilot to see the mark.

For more brevity terms reference [Brevity](#)

Laser Designator Procedures (Include Laser Code in Briefing):

Laser designators guide laser-guided munitions to the target. There are two types of laser designating:

1. **Laser Handoff:**
 - The laser marks the target. Once the pilot has visual confirmation ("tally") on the target, they request the laser to cease. The pilot then fires using their own laser.
2. **Buddy Lasing:**
 - The JTAC maintains laser designation throughout the engagement. The pilot does not need to maintain visual on the target after firing, which is useful when surface-to-air threats are

present. Always include the callsign of the person lasing in the briefing.

In both scenarios, the pilot must be informed of the Laser Target Line (LTL), which is the direction the laser is pointing.

Smoke: (DO NOT USE WHITE SMOKE), mark is smoke (do not say what colour the smoke is) when the pilot gets tally on target he will confirm smoke colour.

5.36 Remarks / Restrictions

JTAC can put any further instructions in here. Here are the most important ones:

- **Final attack heading**, this is the heading in degrees that the attacking aircraft should maintain whilst running into the target. Always say the heading in 3 numbers (i.e. heading 010 or 350). The pilot has around a 20-30 degree leniency unless otherwise advised.
- **LTL / PTL**, use magnetic degrees or cardinal directions.
- **Danger close and initials**, always give initials of the commander approving the danger close mission.

5.36 Initial Points (IP), Battle Points (BP) and Holding Areas (HA) / Holding Points (HP)

Initial Point (IP):

An Initial Point (IP) is a location from which a fixed-wing aircraft begins its run towards the target. IPs can be established before an operation, which is useful for multiple target missions, or set up using the Keyhole method.

IPs are used to:

- Assist the pilot in positioning before the attack.
- Allow the JTAC to direct the aircraft's flight path before weapons release.

Battle Positions (BP)

A Battle Position is a three dimensional space where an aircraft can deploy munitions onto a specified target. The Battle Position can be used as a start point for the aircraft to push to the target or as a stationary position of fire.

Battle Positions are used for rotary wing aircraft, can be pre-planned or setup using the Keyhole Method.

Holding Areas (HA) / Holding Points (HP)

Three dimensional space where aircraft can hold position. Often used before IP/BP locations.

Holding areas are specific places where the aircraft should continue to hold within while Holding Points is a point where the aircraft needs to hold around.

Keyhole Method:

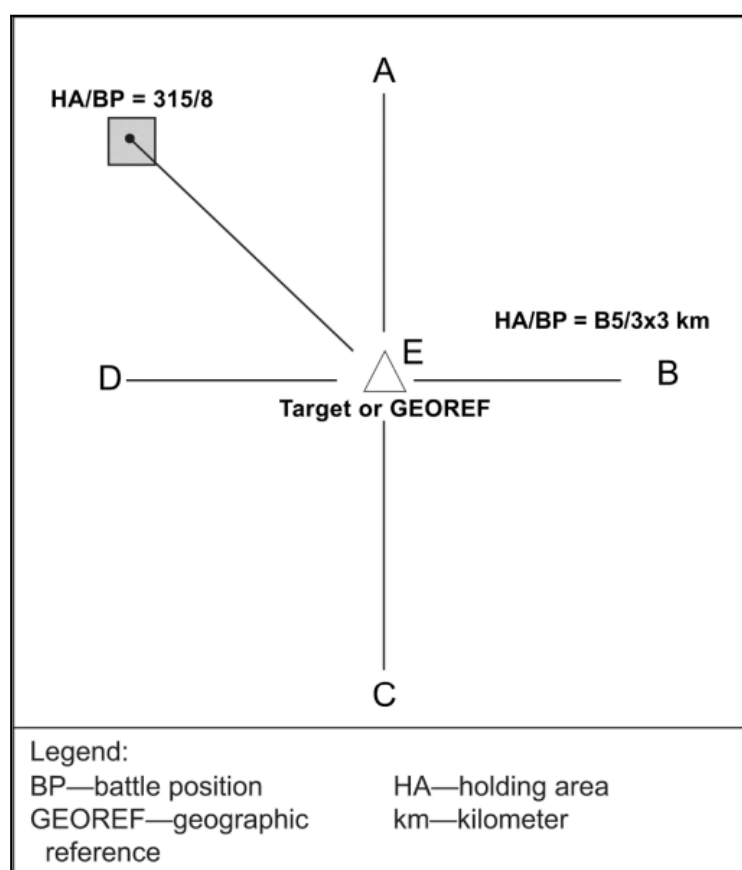
The keyhole template below is an efficient method for establishing an IP/BP/holding area (HA) in the absence of control points or when the control point location does not sufficiently support target engagement.

When using the keyhole template with FW assets, distance represents kilometers from the echo point and closest allowable hold range to the echo point. The target location may be used as the echo point.

When using the keyhole template with RW assets, distance represents km to the center of the battle position (BP), with the standard BP being 300 m x 300 m.

Stack owners may direct CAS aircraft to loiter around the echo point, outside a specified distance, using the term “maintain Echo (distance)” (e.g., “maintain Echo 5”). Can also be referred to as “maintain overhead (grid, distance)”.

Figure below provides an example of the Keyhole Method.



5.4 Intelligence, Surveillance and Recognizance (ISR)

ISR directs an aircraft to collect intelligence, surveillance, and reconnaissance over a specified area. It is often used to provide overwatch for ground troops and detect incoming threats. The aircraft will generate spotting reports (spotreps) to relay the intelligence gathered from the ISR point.

5.5 9-Line Evacuation Request

Also known as a Medevac or Casevac. Used when a patient is in a condition not maintainable within combat and evacuation is needed. When setting up the brief the medic plays a big role in giving the: number of patients, urgency of patients and nationality of patients.

Figure below provides the template of a 9-Line Evacuation Request.

Table 82. 9-Line Evacuation Request

Line	Item	Explanation
1	Location of pickup.	Grid coordinates using the military grid reference system (commonly referred to as MGRS) of the pickup site should be sent by secure communications. To prevent confusion, the grid zone letters are included in the message. Follow standard operating procedure for encryption of grid if using unsecure communications.
2	Radio frequency, call sign, and suffix (if used).	Frequency of the radio at the pickup site, not a relay frequency. The call sign of the person to be contacted at the pickup site may be transmitted in the clear.
3	Number of patients by precedence.	A—Urgent. B—Urgent-surgery. C—Priority. D—Routine. E—Convenience. If two or more categories must be reported in the same request, insert the word "BREAK" between each category.
4	Special equipment required.	A—None. B—Hoist. C—Extraction equipment. D—Ventilator.
5	Number of patients by type.	Litter + number of patients (Litter). Ambulatory + number of patients (Ambulatory).
6	(Wartime) Security of pickup site	N—No enemy in the area. P—Possible enemy in the area, use caution. E—Enemy in the area, approach with caution. X—Enemy in the area, armed escort required.
	(Peacetime) Number and type of wound, injury, or illness.	Specific information regarding patient wounds by type (ex: gunshot or shrapnel). Report serious bleeding, along with patient blood type if known.
7	Method of markings pickup site.	A—Panels. B—Pyrotechnic signal. C—Smoke (color). D—None. E—Other.
8	Patient nationality and status.	The number of patients in each category need not be transmitted. A—United States (US) military B—US citizen C—Non-US military D—Non-US citizen E—Enemy prisoner of war
9	(Wartime) Contamination (only if required).	C—Chemical. B—Biological. R—Radiological. N—Nuclear.
	(Peacetime) Terrain description.	Describe obstacles and prominent features around the landing site (e.g., lake, tower).

5.6 Danger Close parameters

Generally anything within 200-300 meters of friendlies is regarded as Danger Close, although certain munitions can change this value. When doing fire missions the JTAC or FAC(A) should always consider the blast radius, shrapnel radius and cover the friendlies have. If it falls under the expected safety distance the JTAC or FAC(A) should notify the aircrew of the strike being Danger Close.

Keep in mind munitions used by Fixed Wing (FW) will have further danger close ranges than Rotary Wings (RW). Also keep in mind the further a aircraft is when deploying dumb fire weapons the more inaccurate these will be.

Reference for FW munitions:

GBU-12 Paveway II 500lb | **270 meters**

GBU-54 500lb | **270 meters**

GBU-32 1000lb | **340 meters**

AGM-65 Maverick | **175 meters**

Gun 20mm | **100 meters**

Gun 25mm or 30mm | **100 meters**

AC-130 30mm | **100 meters**

AC-130 105mm | **155 meters**

Figure below provides a few distances for certain munitions mainly used by RW.

Table 93. Rotary-wing REDs				
Airframe/ Weapon	Firing Range (meters)	0.1% Probability of Incapacitation (meters)		
		DANGER CLOSE/ Standing	Prone	Prone Protected
AH-1/ 20 mm (M56) ³	300	45	35	35
	800	95	80	80
	1,500	155	140	140
AH-64/ 30 mm (M789) ³	500	55	50	50
	1,000	70	65	60
	1,500	100	90	90
2.75 inch HE rockets (M151/229) ³	300	110	95	90
	800	135	115	110
	1,500	170	155	150
2.75 inch flechette (M255) ³	1,000	235	235	235
AGR-19/20 APKWS	All	105	70	65
AGM-114 FA/K/L/ M/N(4/6)/P(2A)	All	115	95	80
AGM-114R	All	130	100	90
AGM-114R2	All	145	120	110
AGM-114R9E contact (airburst)	All	195 (230)	140 (185)	125 (160)
AGM-114R9H contact (airburst)	All	155 (200)	110 (150)	100 (130)
AGM-179A JAGM contact (airburst)	All	110 (145)	80 (105)	75 (100)
Notes: 1. Door gun risk-estimate distances (REDs) are not available due to the inability to model crew-served weapon employment without a fire control system. 2. The .50 caliber, fixed forward platforms are not able to be modeled at this time. REDs will be released as rotary-wing delivery accuracy programs increase in fidelity for systems without fire control computers. 3. Over the shoulder methodology used for guns and unguided rockets. 4. All Ungrayed systems and P _i are unclassified unless specifically marked as (CUI)				
Legend: APKWS—Advanced Precision Kill Weapon System CUI—controlled unclassified information HE—high explosive JAGM—joint air-to-ground missile mm—millimeter P _i —probability of incapacitation				

Module 6: Air Assault

6.1 Helicopter Landing Zone (HLZ)

Reference  *TFW - Air Assault* and *JTAC Cards*

HLZs are landing zones that are set up by the 18E and leadership. They are used to move personnel and equipment around the Area of Operations (AO). When setting up a HLZ keep in mind the pilot's situation, any AA in the area and the surroundings of the HLZ. Meaning that if the pilot is in a high stress environment such as a hot HLZ then the HLZ should be bigger to allow for error. If there is no contact expected you can narrow the HLZ down more.

The leadership (18Z, 18A or 18F) has final say on a HLZ. The HLZ can be set up remotely, or on site.

6.11 Air Assault Procedures

- 1) JTAC coordinates the HLZ with the pilots. But leadership has final say on the location.
- 2) Leadership coordinates within the helicopter when it comes to: “All in”.JTAC coordinates when the helicopter can take off or touch down.
- 3) Ground troops entering a helicopter will switch their intercom to one of the misc channels. The only people on the crew or cargo channel are the pilots, crew and the JTAC or leadership.

Revision Number	Date	Description of Change	Author/Editor	Approval
1.0	06/08/2024	Initial release	P. Hamilton	N/A
1.1	31/08/2024	- Added Figures - Added Evacuation Requests - Updated Module 5 - Added 5.6 Danger Close parameters	P. Hamilton	N/A